Evan Chilek

chilekevan@gmail.com | (570)-536-2146 | Portfolio: echilek.github.io

Objective

Junior mechanical engineering student in the Schreyer Honors College looking for an internship for Summer 2027. Experience in CAD modeling, computational analysis, and manufacturing through projects and as a mechanical intern designing building systems and creating technical documentation.

Education

The Pennsylvania State University, University Park, PA — GPA: 3.97

B.S. in Mechanical Engineering with honors — Expected: May 2028

Engineering & Employment History

Mechanical Intern, Greenman-Pedersen Inc., Scranton, PA — 5/25-8/25; 5/26-8/26

- Performed heating and cooling load calculations using Trace 700 to size HVAC equipment and develop mechanical systems, ensuring compliance with performance and ASHRAE standards
- Developed detailed mechanical drawings, BIM models, and created schedules in Revit and AutoCAD for commercial and institutional building projects

Cashier, Wegmans, Dickson City, PA — 11/20-Present

Engineering Projects

Mu-Bot Research, BioRob InFL Lab — Fall 2026

- Undergraduate research assistant on a project with the goal of developing autonomous robotic swimmer
- Exploring and testing how different materials and mechanical designs affect the elasticity and modularity of the body to maximize performance in bursts and efficiency in coasting

Small-Scale Motorjet Propulsion System, Personal Project — Summer 2026

- Designed a small-scale motorjet around a 70 mm electric ducted fan, developing the casing, structural supports, combustion chamber, and subsystem layout in Fusion 360
- Built and independently verified the electrical/control and ignition subsystems, confirming their operation before structural integration of the full motorjet assembly
- Fabricated thin-gauge steel components and evaluated forming, joining, assembly, and manufacturability constraints encountered during the first build

Enterprise Drone, Manufacturing Lead, ASME Design Team — Fall 2025- Spring 2026

- Performed testing and conducted finite element analysis on Fusion 360 to drive iterative design improvements to increase performance by reducing weight and maintain a factor of safety
- Selected materials such as CFRP, PLA, and aluminum by performing trade-off analysis considering mechanical properties, cost, and manufacturability to support design decisions

Technical Skills

Certification: Autodesk Certified Associate in CAD for Mechanical Design — ADSK-159-1196

Modeling/Analysis: Fusion 360, SolidWorks, AutoCAD, Revit/MATLAB, Fusion 360(simulation)

Manufacturing/Prototyping: 3D printing, abrasive waterjet cutting, carbon fiber fabrication, woodworking/joinery, mechanical assembly

Awards

Wabtec Design Competition: Most Sustainable Design